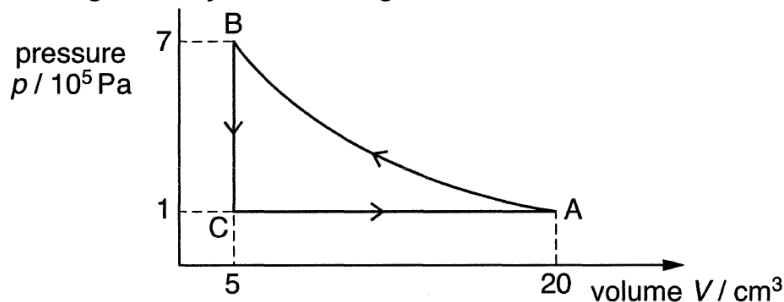


- 11 An ideal gas undergoes a cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$, as shown.



Which of the following statements is correct?

- A Process $A \rightarrow B$ is isothermal.
 - B Process $B \rightarrow C$ is isochoric.
 - C Work of 15 J is done by the gas in process $C \rightarrow A$.
 - D The internal energy of the gas is zero.
- B** constant volume process is also known as isochoric process,
Option A is wrong as pV is not constant which would be the case if T is constant (isothermal).
Option C is a trick question to see if students noticed the prefix in the V axis. V is in cm^3 while P is of the order of magnitude of 5.
Option D is a trick on semantics. While the CHANGE in internal energy of the gas may be zero in this cycle, the internal energy is not zero.

12 Fig. 12.1 shows the variation with displacement x of velocity of a simple harmonic